UNIVERSIDAD DE LAS FUERZAS ARMADAS – ESPE
Department of Computer Science
Software Engineering
36862 – Object-Oriented Programming

UNIT 1 PROJECT · TEAM WORK

Test Case Specification: ProvStock

Prov&Tec Management, a Comprehensive Sales and Inventory System

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Revision Note

> *Version 1.1 updates the test cases after the changes to the program: the data is now saved in JSON files, one per class; the reports are exported to CSV files; and the code was reviewed with the book Clean Code, so every error message now starts with "Error:". All the test cases were executed again on October 26, 2026. TC-33 and TC-34 were rewritten for the CSV reports and the JSON files, the actual results of TC-10, TC-11, TC-15, and TC-26 were updated with the new messages, and the observations of version 1.0 were closed (Section 4). The test cases come from the Use Case Specifications v1.2 and the Software Requirements Specification (SRS) v1.3.*

1. Introduction

1.1 Purpose

This document describes the test cases used to check that ProvStock, the sales and inventory management system for Prov&Tec, does what the requirements say. Each test case gives the data to enter, the steps to follow in the console menu, the result that the system should show, and the result that it actually showed when the team ran the test.

1.2 Scope

The document has 34 functional test cases, written from the point of view of the user (black-box tests): 20 positive cases, which follow the main flow with valid data, and 14 negative cases, which enter invalid data or follow an exception path. Together they cover the 20 use cases and the nine Product Backlog items, plus data persistence.

The tests that need special conditions are outside the scope of this version: performance with large volumes of data (NFR-01), usability with real users (NFR-03), and installation on different computers (NFR-06). The tests of individual classes and methods are the unit tests described in Section 5.

1.3 Conventions

Every test case is described with the same template (Table 1). The requirement identifiers (FR, NFR, BR, UI, SI) refer to the SRS v1.3, and the use case identifiers (UC) refer to the Use Case Specifications v1.2. The texts in quotation marks are the messages that the program shows in Spanish.

Table 1
Test Case Template

Field	Content
Test case	Identifier (TC-xx) and name of the test.
Use case	The use case and, in parentheses, the step or exception path that is tested.
Requirements	The SRS requirements and business rules that the test case checks.
Type	Positive (valid data and main flow) or Negative (invalid data or exception path).
Preconditions	The state of the system and the user who must be logged in before the test.
Test data	The exact values entered in the console.
Steps	The numbered actions that the tester performs.
Expected result	What the system must do according to the requirements.
Actual result	What the system did when the test was executed.
Status	Passed, if the actual result matches the expected result; otherwise, Failed.

1.4 Test Environment

Table 2 describes the environment in which the test cases were executed.

Table 2
Test Environment

Item	Value
System under test	ProvStock, console version (package ec.edu.espe.provstock), main class view.ProvStockApp.
Java	Java Development Kit 21.
Data	Folder data of the project, with one JSON file per class. It was deleted before each test case, so every test starts from the same sample data.
Reports	Folder reports of the project, where the CSV files of the reports are saved.
Execution date	October 26, 2026.
Testers	Team ByteWorks.

1.5 Test Data

The first time the program runs, it creates the sample data of Tables 3 to 5. Two products start at or below their minimum stock (CAB-002 and PRO-001), so the warehouse clerk and the owner see two stock alerts from the beginning.

Table 3
Test Users

Username	Password	Full name	Role
owner	Owner2026	Propietario	Owner
vendedor	Vendedor2026	María López	Salesperson
bodeguero	Bodega2026	Pedro Ramírez	Warehouse Clerk

Table 4
Test Products

Code	Name	Unit	Stock	Minimum	Cost	Price
CAB-001	Cable THHN 12 AWG	Meter	500	100	0.45	0.70
CAB-002	Cable gemelo 2x14 AWG	Meter	80	100	0.35	0.55
ILU-001	Foco LED 9W luz blanca	Unit	60	20	1.20	2.50
INT-001	Interruptor simple	Unit	40	15	1.10	2.00
PRO-001	Breaker 1 polo 20A	Unit	8	10	4.50	7.80
HER-001	Alicate universal 8 pulgadas	Unit	12	5	6.00	10.50

Note. Costs and prices are in U.S. dollars without VAT. The initial stock is loaded with a stock adjustment whose reason is “Inventario inicial”.

Table 5
Test Customers and Suppliers

Type	Identification	Name
Customer	9999999999999 (RUC)	CONSUMIDOR FINAL (created by the system)
Customer	1712345675 (cédula)	Juan Pérez
Customer	1701234567001 (RUC)	Instalaciones Andes
Supplier	1790012345001	Electrocables S.A.
Supplier	0992345678001	Distribuidora Eléctrica del Pacífico

2. Test Case Summary

Table 6 lists the 34 test cases with the use case they come from, their type, and their status after the execution.

Table 6
Test Case Summary

ID	Test case	Use case	Type	Status
TC-01	Log in with valid credentials	UC-01	Positive	Passed
TC-02	Log in with a wrong password	UC-01	Negative	Passed
TC-03	Log in with a deactivated account	UC-04	Negative	Passed
TC-04	Main menu shows only the functions of the role	UC-01	Positive	Passed
TC-05	Change own password	UC-03	Positive	Passed
TC-06	Change password with fewer than 8 characters	UC-03	Negative	Passed
TC-07	Create a user account	UC-04	Positive	Passed
TC-08	Create a user with a username that already exists	UC-04	Negative	Passed
TC-09	Register a customer with a valid cédula	UC-05	Positive	Passed
TC-10	Register a customer with a duplicate identification	UC-05	Negative	Passed
TC-11	Register a customer with an invalid cédula	UC-05	Negative	Passed
TC-12	Try to deactivate the Final Consumer	UC-05	Negative	Passed
TC-13	Search products by name	UC-06	Positive	Passed
TC-14	Register a new product	UC-07	Positive	Passed
TC-15	Register a product with a duplicate code	UC-07	Negative	Passed
TC-16	Register a sale to the Final Consumer	UC-08, UC-11	Positive	Passed
TC-17	Sell a quantity greater than the stock	UC-08	Negative	Passed
TC-18	Void a registered sale	UC-09, UC-11	Positive	Passed
TC-19	Void a sale that is already voided	UC-09	Negative	Passed
TC-20	Update the VAT rate	UC-10, UC-16	Positive	Passed
TC-21	Update the VAT rate with an invalid value	UC-10	Negative	Passed

ID	Test case	Use case	Type	Status
TC-22	View the inventory and the movements of a product	UC-12	Positive	Passed
TC-23	Adjust the stock with a reason	UC-13	Positive	Passed
TC-24	Adjust the stock to a negative value	UC-13	Negative	Passed
TC-25	Register a supplier	UC-14	Positive	Passed
TC-26	Register a supplier with an invalid RUC	UC-14	Negative	Passed
TC-27	Register a purchase that removes a stock alert	UC-15, UC-11, UC-17	Positive	Passed
TC-28	Filter the sales history by status	UC-16	Positive	Passed
TC-29	View the active stock alerts	UC-18, UC-17	Positive	Passed
TC-30	A sale creates a new stock alert	UC-08, UC-11, UC-17	Positive	Passed
TC-31	Generate the sales report	UC-19	Positive	Passed
TC-32	Generate a report with the start date after the end date	UC-19	Negative	Passed
TC-33	Export the inventory report to a CSV file	UC-19, UC-20	Positive	Passed
TC-34	Data is kept after closing the program	All	Positive	Passed

3. Test Cases

Each test case is described below with the template of Table 1, grouped by Product Backlog item.

3.1 Access Management (RF01)

TC-01. LOG IN WITH VALID CREDENTIALS	
Use case	UC-01 Log in
Requirements	FR-01.2, FR-08.1, UI-02
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5).
Test data	Username: owner · Password: Owner2026
Steps	1. Run the program and choose option 1 (Iniciar sesión). 2. Enter the username and the password. 3. Look at the main menu.
Expected result	The system shows “Bienvenido/a, Propietario (OWNER)”, warns that 2 products have low stock, and shows the main menu with the 15 options of the owner and the counter of stock alerts.
Actual result	The welcome message, the warning “Hay 2 producto(s) con stock bajo”, and the 15 options of the owner were shown.
Status	Passed

TC-02. LOG IN WITH A WRONG PASSWORD	
Use case	UC-01 Log in (3a)
Requirements	FR-01.3
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5).
Test data	Username: owner · Password: clave123
Steps	1. Choose option 1 (Iniciar sesión). 2. Enter the username and the wrong password.
Expected result	The login is rejected with a general message that does not say which of the two values is wrong, and the start menu is shown again.
Actual result	The system showed “Usuario o contraseña incorrectos, o la cuenta está inactiva.” and returned to the start menu.
Status	Passed

TC-03. LOG IN WITH A DEACTIVATED ACCOUNT	
Use case	UC-04 Manage user accounts (A2), UC-01 (3a)
Requirements	FR-01.4, BR-07
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	Account to deactivate: vendedor
Steps	1. Open “Gestionar usuarios” and choose “Desactivar usuario”. 2. Enter vendedor and confirm with s. 3. Log out. 4. Try to log in as vendedor / Vendedor2026.
Expected result	The account is deactivated, and the deactivated user cannot log in even with the correct password.
Actual result	“Usuario desactivado.” was shown; the later login was rejected with the general error message.
Status	Passed

TC-04. MAIN MENU SHOWS ONLY THE FUNCTIONS OF THE ROLE	
Use case	UC-01 Log in (step 4)
Requirements	FR-01.1, NFR-02, UI-02
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5).
Test data	Username: vendedor · Password: Vendedor2026
Steps	1. Log in as the salesperson. 2. Read the options of the main menu.
Expected result	Only the salesperson options appear: search products, manage customers, register sale, and change password. There is no alert counter.
Actual result	The menu showed exactly 4 options (Buscar productos, Gestionar clientes, Registrar venta, Cambiar mi contraseña) and no stock alerts.
Status	Passed

TC-05. CHANGE OWN PASSWORD	
Use case	UC-03 Change password
Requirements	FR-01.5, NFR-02
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as vendedor / Vendedor2026.
Test data	Current: Vendedor2026 · New and confirmation: NuevaClave26
Steps	1. Choose “Cambiar mi contraseña”. 2. Enter the current password, the new password, and its confirmation. 3. Log out and try to log in with the old password. 4. Log in with the new password.
Expected result	The password is changed; after that, the old password is rejected and the new one is accepted.
Actual result	“Contraseña cambiada correctamente.” was shown; the old password was rejected and the new one opened the session.
Status	Passed

TC-06. CHANGE PASSWORD WITH FEWER THAN 8 CHARACTERS	
Use case	UC-03 Change password (4b)
Requirements	NFR-02
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as vendedor / Vendedor2026.
Test data	Current: Vendedor2026 · New and confirmation: abc12
Steps	1. Choose “Cambiar mi contraseña”. 2. Enter the current password and a new password of 5 characters twice.
Expected result	The change is rejected with an error message, and the previous password is kept.
Actual result	“No se cambió la contraseña: … la nueva tiene menos de 8 caracteres …” was shown.
Status	Passed

TC-07. CREATE A USER ACCOUNT	
Use case	UC-04 Manage user accounts
Requirements	FR-01.4, NFR-02
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	Username: jgomez · Name: Juan Gómez · Role: SALESPERSON · Password: Clave2026
Steps	1. Open “Gestionar usuarios” and choose “Crear usuario”. 2. Enter the data and choose the role 1 (SALESPERSON). 3. Choose “Listar usuarios”. 4. Log out and log in as jgomez / Clave2026.
Expected result	The account is created as active, appears in the list with its role, and the new user can log in.
Actual result	“Usuario creado: jgomez (id 4)” was shown, the list showed it as Activo, and the login as Juan Gómez worked.
Status	Passed

TC-08. CREATE A USER WITH A USERNAME THAT ALREADY EXISTS	
Use case	UC-04 Manage user accounts (4a)
Requirements	BR-08
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	Username: vendedor (already exists) · Password: Clave2026
Steps	1. Open “Gestionar usuarios” and choose “Crear usuario”. 2. Enter the existing username and the other data.
Expected result	The account is not created, and an error message says that the username already exists.
Actual result	“Error: Ya existe el usuario vendedor” was shown, and no account was created.
Status	Passed

3.2 Customer Management (RF02)

TC-09. REGISTER A CUSTOMER WITH A VALID CÉDULA	
Use case	UC-05 Manage customers
Requirements	FR-02.1, FR-02.2, FR-02.4
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as vendedor / Vendedor2026.
Test data	Type: CEDULA · Number: 1720456787 · Name: Carlos Andrade · Phone: 0987654321 · Email: carlos.andrade@mail.com · Address: Sangolquí
Steps	1. Open “Gestionar clientes” and choose “Registrar cliente”. 2. Enter the data. 3. Choose “Buscar clientes” and search for Andrade.
Expected result	The customer is saved as active and is found by part of the name.
Actual result	“Cliente registrado con id 4.” was shown, and the search returned Carlos Andrade with his cédula.
Status	Passed

TC-10. REGISTER A CUSTOMER WITH A DUPLICATE IDENTIFICATION	
Use case	UC-05 Manage customers (3a)
Requirements	FR-02.2, BR-08
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as vendedor / Vendedor2026.
Test data	Type: CEDULA · Number: 1712345675 (belongs to Juan Pérez) · Name: Ana Ruiz
Steps	1. Open “Gestionar clientes” and choose “Registrar cliente”. 2. Enter the identification number that already exists.
Expected result	The customer is not saved, and the message says that the identification already exists.
Actual result	“Error: Ya existe un cliente con la identificación 1712345675.” was shown.
Status	Passed

TC-11. REGISTER A CUSTOMER WITH AN INVALID CÉDULA	
Use case	UC-05 Manage customers (3b)
Requirements	FR-02.2
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as vendedor / Vendedor2026.
Test data	Type: CEDULA · Number: 1712345678 (wrong check digit) · Name: Ana Ruiz
Steps	1. Open “Gestionar clientes” and choose “Registrar cliente”. 2. Enter the cédula with the wrong last digit.
Expected result	The customer is not saved because the cédula does not pass the modulo 10 check.
Actual result	“Error: La identificación no es válida para el tipo CEDULA.” was shown.
Status	Passed

TC-12. TRY TO DEACTIVATE THE FINAL CONSUMER	
Use case	UC-05 Manage customers (A3)
Requirements	FR-02.5, FR-02.6
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	Identification: 9999999999999
Steps	1. Open “Gestionar clientes” and choose “Desactivar cliente”. 2. Enter 9999999999999 and confirm with s.
Expected result	The system does not allow it, and the Final Consumer stays active.
Actual result	“Error: El Consumidor Final no puede desactivarse” was shown.
Status	Passed

3.3 Product Management (RF03)

TC-13. SEARCH PRODUCTS BY NAME	
Use case	UC-06 Search products
Requirements	FR-03.4
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as vendedor / Vendedor2026.
Test data	Text: cable · Category: 0 (all)
Steps	1. Choose “Buscar productos”. 2. Enter the text and choose all the categories.
Expected result	Only the products whose name contains “cable” are listed, with their price, stock, and location.
Actual result	CAB-001 (500.00 m, $0.70, Pasillo A1) and CAB-002 (80.00 m, $0.55, Pasillo A2, marked “stock bajo”) were listed.
Status	Passed

TC-14. REGISTER A NEW PRODUCT	
Use case	UC-07 Manage products
Requirements	FR-03.1, FR-03.2, FR-05.4
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as bodeguero / Bodega2026.
Test data	Code: TOM-001 · Name: Tomacorriente doble · Category: SWITCHES_AND_OUTLETS · Brand: Veto · Unit: UNIT · Location: Pasillo C2 · Cost: 1.30 · Price: 2.40 · Minimum: 10 · Initial stock: 25
Steps	1. Open “Gestionar productos” and choose “Registrar producto”. 2. Enter the data. 3. Choose “Listar productos activos”.
Expected result	The product is saved as active and appears in the list with its initial stock, which is loaded as an adjustment.
Actual result	“Producto registrado con id 7.” was shown, and TOM-001 appeared in the list with 25.00 units.
Status	Passed

TC-15. REGISTER A PRODUCT WITH A DUPLICATE CODE	
Use case	UC-07 Manage products (3a)
Requirements	FR-03.2, BR-08
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as bodeguero / Bodega2026.
Test data	Code: CAB-001 (already exists)
Steps	1. Open “Gestionar productos” and choose “Registrar producto”. 2. Enter the existing code.
Expected result	The system rejects the code before asking for the other data.
Actual result	“Error: Ya existe un producto con el código CAB-001.” was shown right after the code was entered.
Status	Passed

3.4 Sales Management (RF04)

TC-16. REGISTER A SALE TO THE FINAL CONSUMER	
Use case	UC-08 Register sale, UC-11 Update stock
Requirements	FR-04.1 to FR-04.7, FR-05.1, BR-05, BR-06
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as vendedor / Vendedor2026.
Test data	Customer: Enter (Final Consumer) · Product: ILU-001 · Quantity: 4 · Payment: CASH
Steps	1. Choose “Registrar venta” and press Enter to sell to the Final Consumer. 2. Choose “Agregar producto”, search ILU-001, and enter 4. 3. Choose “Confirmar venta”, select CASH, and confirm with s. 4. Log in as bodeguero and open “Consultar inventario” for ILU-001.
Expected result	Subtotal $10.00, VAT 15% $1.50, and total $11.50. The sale gets number 1, and the stock of ILU-001 goes from 60 to 56 with a SALE movement.
Actual result	The detail showed 10.00 / 1.50 / 11.50 and “Venta Nº 1 registrada correctamente.”; the inventory showed 56.00 and the movement SALE −4.00 “Venta Nº 1”.
Status	Passed

TC-17. SELL A QUANTITY GREATER THAN THE STOCK	
Use case	UC-08 Register sale (4a)
Requirements	FR-04.4, BR-03
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as vendedor / Vendedor2026.
Test data	Product: HER-001 (stock 12) · Quantity: 20
Steps	1. Choose “Registrar venta” and press Enter. 2. Choose “Agregar producto”, search HER-001, and enter 20. 3. Cancel the sale.
Expected result	The line is rejected, the available stock is shown, and nothing is saved.
Actual result	“Error: Stock insuficiente de Alicate universal 8 pulgadas. Disponible: 12.0” was shown; the sale was cancelled.
Status	Passed

TC-18. VOID A REGISTERED SALE	
Use case	UC-09 Void sale, UC-11 Update stock
Requirements	FR-04.8, FR-05.3, BR-04
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Sale Nº 1 registered by vendedor (ILU-001 × 4, as in TC-16). Logged in as owner / Owner2026.
Test data	Sale number: 1 · Reason: Error de digitación
Steps	1. Choose “Anular venta” and enter 1. 2. Enter the reason and confirm with s. 3. Open “Consultar inventario” for ILU-001.
Expected result	The sale changes to VOIDED with the reason, date, and user, and the stock of ILU-001 returns to 60 with a VOIDED_SALE movement.
Actual result	“Venta anulada y stock restaurado.” was shown; the stock returned to 60.00 and the movement VOIDED_SALE +4.00 appeared.
Status	Passed

TC-19. VOID A SALE THAT IS ALREADY VOIDED	
Use case	UC-09 Void sale (1a)
Requirements	FR-04.8
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Sale Nº 1 already voided (as in TC-18). Logged in as owner / Owner2026.
Test data	Sale number: 1
Steps	1. Choose “Anular venta” and enter 1.
Expected result	The sale is shown with its void data, and the system does not let the owner void it again.
Actual result	The sale showed “ANULADA el 26/10/2026 … Motivo: Error de digitación” and the message “La venta ya está anulada.”
Status	Passed

TC-20. UPDATE THE VAT RATE	
Use case	UC-10 Update VAT rate, UC-16
Requirements	FR-04.9, BR-05
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Sale Nº 1 registered at 15% (ILU-001 × 4). Logged in as owner / Owner2026.
Test data	New rate: 12 · New sale: ILU-001 × 4
Steps	1. Choose “Actualizar tasa de IVA”, enter 12, and confirm. 2. Register a new sale of ILU-001 × 4. 3. Open “Historial de ventas” without filters.
Expected result	The new sale uses 12% (total $11.20), and sale Nº 1 keeps 15% (total $11.50).
Actual result	“Tasa de IVA actualizada a 12%.” was shown; sale Nº 2 had VAT 12% $1.20 and total $11.20, while sale Nº 1 kept $11.50.
Status	Passed

TC-21. UPDATE THE VAT RATE WITH AN INVALID VALUE	
Use case	UC-10 Update VAT rate (3a)
Requirements	FR-04.9
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	New rate: 150
Steps	1. Choose “Actualizar tasa de IVA”, enter 150, and confirm with s.
Expected result	The value is rejected and the rate stays at 15%.
Actual result	“Error: La tasa de IVA debe estar entre 0% y 99%” was shown, and the menu still showed IVA 15%.
Status	Passed

3.5 Inventory Management (RF05)

TC-22. VIEW THE INVENTORY AND THE MOVEMENTS OF A PRODUCT	
Use case	UC-12 View inventory
Requirements	FR-05.5, FR-05.6
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as bodeguero / Bodega2026.
Test data	Filter: Enter · Category: 0 · Product: CAB-001
Steps	1. Choose “Consultar inventario” without filters. 2. Enter CAB-001 to see its movements.
Expected result	All the active products are listed with their stock, and the products at or below the minimum are highlighted. CAB-001 shows its initial inventory movement.
Actual result	The 6 products were listed, CAB-002 and PRO-001 marked “(stock bajo)”; CAB-001 showed ADJUSTMENT +500.00 “Inventario inicial”.
Status	Passed

TC-23. ADJUST THE STOCK WITH A REASON	
Use case	UC-13 Adjust stock
Requirements	FR-05.4
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as bodeguero / Bodega2026.
Test data	Product: HER-001 (stock 12) · Quantity: 5 · Reason: Conteo físico: 5 unidades sin registrar
Steps	1. Choose “Ajustar stock” and enter HER-001. 2. Enter the quantity and the reason.
Expected result	The stock of HER-001 changes from 12 to 17, and an ADJUSTMENT movement is saved with the user and the reason.
Actual result	“Stock ajustado. Nuevo stock: 17.0” was shown.
Status	Passed

TC-24. ADJUST THE STOCK TO A NEGATIVE VALUE	
Use case	UC-13 Adjust stock (4a)
Requirements	BR-03
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as bodeguero / Bodega2026.
Test data	Product: PRO-001 (stock 8) · Quantity: −20 · Reason: Producto dañado
Steps	1. Choose “Ajustar stock” and enter PRO-001. 2. Enter −20 and the reason.
Expected result	The adjustment is rejected because the stock would be negative, and the stock stays at 8.
Actual result	“Error: El stock de Breaker 1 polo 20A no puede quedar negativo” was shown.
Status	Passed

3.6 Supplier and Purchase Management (RF06)

TC-25. REGISTER A SUPPLIER	
Use case	UC-14 Manage suppliers
Requirements	FR-06.1, FR-06.2
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	RUC: 1791234567001 · Name: Iluminación Total Cía. Ltda. · Contact: Rosa Vega · Phone: 022555666 · Email: ventas@iluminaciontotal.com · Address: Quito
Steps	1. Open “Gestionar proveedores” and choose “Registrar proveedor”. 2. Enter the data. 3. Choose “Buscar proveedores” and search for Iluminación.
Expected result	The supplier is saved as active and is found by its name.
Actual result	“Proveedor registrado con id 3.” was shown, and the search returned the supplier with its RUC and contact.
Status	Passed

TC-26. REGISTER A SUPPLIER WITH AN INVALID RUC	
Use case	UC-14 Manage suppliers (3a)
Requirements	FR-06.1
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	RUC: 1791234567000 (ends in 000) · Name: Proveedor Falso · Phone: 022555666
Steps	1. Open “Gestionar proveedores” and choose “Registrar proveedor”. 2. Enter the data with the invalid RUC.
Expected result	The supplier is not saved, and a message says that the RUC is not valid.
Actual result	“Error: El RUC no es válido.” was shown.
Status	Passed

TC-27. REGISTER A PURCHASE THAT REMOVES A STOCK ALERT	
Use case	UC-15 Register purchase, UC-11, UC-17
Requirements	FR-06.3 to FR-06.5, FR-05.2, FR-08.4
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as bodeguero / Bodega2026.
Test data	Supplier: 1790012345001 · Invoice: 001-001-000123 · Product: CAB-002 (stock 80, minimum 100) · Quantity: 50 · Unit cost: Enter (0.35)
Steps	1. Choose “Registrar compra” and enter the RUC and the invoice number. 2. Choose “Agregar producto”, search CAB-002, enter 50, and keep the cost. 3. Choose “Confirmar compra” and confirm with s. 4. Open “Alertas de stock”.
Expected result	The purchase gets number 1 with a total cost of $17.50. The stock of CAB-002 rises to 130, so its alert disappears and only PRO-001 remains.
Actual result	“Compra Nº 1 registrada. El stock fue actualizado.” was shown with total 17.50; the alert list showed only PRO-001 (1 alert).
Status	Passed

3.7 Sales History (RF07)

TC-28. FILTER THE SALES HISTORY BY STATUS	
Use case	UC-16 View sales history
Requirements	FR-07.1, FR-07.2
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Sales Nº 1 (ILU-001 × 4) and Nº 2 (CAB-001 × 10) registered; sale Nº 1 voided. Logged in as owner / Owner2026.
Test data	Dates, salesperson, and customer: Enter · Status: 2 (VOIDED)
Steps	1. Choose “Historial de ventas”. 2. Leave the first filters empty and choose the status VOIDED.
Expected result	Only sale Nº 1 is listed, with its date, customer, salesperson, status, and total.
Actual result	Only “1 … CONSUMIDOR FINAL vendedor VOIDED 11.50” was listed.
Status	Passed

3.8 Stock Alerts (RF08)

TC-29. VIEW THE ACTIVE STOCK ALERTS	
Use case	UC-18 Receive stock alerts, UC-17
Requirements	FR-08.2, FR-08.3
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as bodeguero / Bodega2026.
Test data	—
Steps	1. Choose “Alertas de stock”.
Expected result	The products at or below their minimum are listed with their stock, minimum, and location: CAB-002 and PRO-001.
Actual result	“Productos con stock igual o menor al mínimo (2)” listed CAB-002 (80/100, Pasillo A2) and PRO-001 (8/10, Pasillo D1).
Status	Passed

TC-30. A SALE CREATES A NEW STOCK ALERT	
Use case	UC-08, UC-11, UC-17 Check minimum stock
Requirements	FR-08.1, FR-08.2
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as vendedor / Vendedor2026.
Test data	Product: INT-001 (stock 40, minimum 15) · Quantity: 30
Steps	1. Register a sale of INT-001 × 30 to the Final Consumer. 2. Log in as bodeguero and open “Alertas de stock”.
Expected result	The sale warns that INT-001 has low stock (10), and the warehouse clerk now sees 3 alerts.
Actual result	“Aviso: Interruptor simple quedó con stock bajo (10.0).” was shown; the alert list had 3 products, including INT-001 (10/15).
Status	Passed

3.9 Reports (RF09)

TC-31. GENERATE THE SALES REPORT	
Use case	UC-19 Generate reports
Requirements	FR-09.1
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Sales Nº 1 (ILU-001 × 4, $11.50) and Nº 2 (CAB-001 × 10, $8.05) registered by vendedor. Logged in as owner / Owner2026.
Test data	Report: 1 (ventas) · From: 2026-10-01 · To: 2026-10-31 · Export: n
Steps	1. Choose “Generar reportes” and the sales report. 2. Enter the dates and answer n to the export question.
Expected result	The report shows 2 sales, total sold $19.55, VAT collected $2.55, the total per salesperson, and the best-selling products.
Actual result	The report showed “Ventas registradas: 2”, “Total vendido: $19.55”, “IVA recaudado: $2.55”, María López $19.55, and CAB-001 and ILU-001 as best sellers.
Status	Passed

TC-32. GENERATE A REPORT WITH THE START DATE AFTER THE END DATE	
Use case	UC-19 Generate reports (2a)
Requirements	FR-09.1
Type	Negative
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	Report: 1 (ventas) · From: 2026-10-31 · To: 2026-10-01
Steps	1. Choose “Generar reportes” and the sales report. 2. Enter the dates in the wrong order.
Expected result	The report is not generated, and an error message explains the problem with the dates.
Actual result	“Error: La fecha de inicio no puede ser posterior a la fecha de fin” was shown.
Status	Passed

TC-33. EXPORT THE INVENTORY REPORT TO A CSV FILE	
Use case	UC-19 Generate reports, UC-20 Export report to CSV
Requirements	FR-09.2, FR-09.4
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	Report: 2 (inventario)
Steps	1. Choose “Generar reportes” and the inventory report. 2. Read the message shown after the report. 3. Open the generated file of the folder reports with Excel.
Expected result	The report shows the 6 products with the value of the stock ($477.00 in total), and a CSV file with the date and time in its name is created in the folder reports with the same data in columns.
Actual result	The report showed “Valor total del stock: $477.00” and “Reporte CSV guardado en: …/reports/reporte_inventario_20261026_134711.csv”; the file opened in Excel with one row per product (6 products) and the total 477.00.
Status	Passed

3.10 Data Persistence

TC-34. DATA IS KEPT AFTER CLOSING THE PROGRAM	
Use case	All (persistence)
Requirements	NFR-04, SI-02
Type	Positive
Preconditions	The folder data was deleted, so the program starts with the sample data (Section 1.5). Logged in as owner / Owner2026.
Test data	Customer: CEDULA 1720456787, Carlos Andrade
Steps	1. Register the customer and exit the program with option 0. 2. Run the program again and log in as owner. 3. Search for the customer 1720456787. 4. Open the folder data.
Expected result	On the second run, the program loads the JSON files of the folder data instead of the sample data, and the customer is found. The folder has one JSON file per class.
Actual result	The second run showed “Datos cargados desde la carpeta "data".”, and the search returned Carlos Andrade. The folder data had the 11 files role.json, user.json, customer.json, product.json, supplier.json, sale.json, saleDetail.json, purchase.json, purchaseDetail.json, stockMovement.json, and settings.json, and customer.json contained the new customer.
Status	Passed

4. Results

Table 7 summarizes the results of the execution. All 34 test cases passed: the program showed the expected messages, calculated the amounts correctly, and kept the stock equal to the sum of its movements.

Table 7
Results per Product Backlog Item

Backlog	Module	Test cases	Passed	Failed
RF01	Access Management	8	8	0
RF02	Customer Management	4	4	0
RF03	Product Management	3	3	0
RF04	Sales Management	6	6	0

Backlog	Module	Test cases	Passed	Failed
RF05	Inventory Management	3	3	0
RF06	Supplier and Purchase Management	3	3	0
RF07	Sales History	1	1	0
RF08	Stock Alerts	2	2	0
RF09	Reports	3	3	0
—	Data Persistence	1	1	0
Total		34	34	0

The three observations of version 1.0 were solved in the current version of the program:

- **TC-33.** The reports are exported to CSV instead of PDF. The file is saved automatically in the folder reports, and it opens in Excel with the same data as the screen.
- **TC-34.** The data is saved in JSON files, one per class, instead of CSV files. The test case now also checks the files of the folder data.
- **Messages.** Every error is now shown with the prefix “Error:” (for example, TC-08, TC-10, TC-11, TC-15, and TC-26), so the messages are consistent (NFR-03).

5. Relation with the Unit Tests

The unit tests check the methods of each class separately with JUnit, while the test cases of this document check the complete flows through the menu. Table 8 relates each test class of the folder src/test to the test cases that cover the same behavior.

Table 8
Unit Test Classes and Related Test Cases

Test class	Class tested	Related test cases
AccessManagerTest	AccessManager	TC-01, TC-02, TC-04, TC-07, TC-08
UserTest	User	TC-03, TC-05, TC-06
RoleTest	Role	TC-04
CustomerManagerTest	CustomerManager	TC-09, TC-10, TC-12
CustomerTest	Customer	TC-09, TC-11, TC-12

Test class	Class tested	Related test cases
IdentificationValidatorTest	IdentificationValidator	TC-11, TC-26
ProductCatalogTest	ProductCatalog	TC-13, TC-14, TC-15
ProductTest	Product	TC-14, TC-24, TC-29
SalesManagerTest	SalesManager	TC-16, TC-20, TC-21
SaleTest	Sale	TC-16, TC-17, TC-18, TC-19
SaleDetailTest	SaleDetail	TC-16
InventoryTest	Inventory	TC-16, TC-18, TC-22, TC-23, TC-24
SupplierTest	Supplier	TC-25, TC-26
PurchaseTest	Purchase	TC-27
PurchaseDetailTest	PurchaseDetail	TC-27

Note. PurchasingManager, StockMonitor, ReportGenerator, FileManager, CsvReportExporter, SaleFilter, and the four report classes do not have unit tests yet; for now, they are checked only by TC-25 to TC-34. ProductInfo is tested through ProductTest. The 103 unit tests did not change in this version: 67 pass and 36 fail on purpose, as the instructor asked.

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